

How drugs act:

general principles

PHARMACODYNAMICS :

the ways in which drugs work or act

PHARMACOKINETICS :

the relationship between dosage and the changes in concentration of drugs in different parts of the body with time

The binding of drug molecules to cells

Drug : **a chemical applied to a physiological system that affect its function in a specific way**

Paul Ehrlich : chemical interactions between drugs and tissues

The drug molecules must exert some chemical influence on one or more constituents of cell in order to produce a pharmacological response.

The drug molecules must be 'bound' to particular constituents of cells and tissues in order to produce an effect.

Ehrlich : **'A drug will not work unless it is bound'.**

Drug targets : Most drugs act on protein molecules.

Protein targets for drug binding

1. **Receptors**
2. **Enzymes**
3. **Carrier molecules (transporters)**
4. **Ion channels**

Drug receptors

- In pharmacology, receptor describes protein molecules those function is to recognize and respond to endogenous chemical signals.

- Receptors form a key part of the system of chemical communication that all multicellular organism use to coordinate the activities of their cells and organs.

- There is an important distinction between ***agonists***, which 'activate' the receptors, and ***antagonists***, which may combine at the same site without causing activation, and block the effect of agonists on that receptor.

Drug specificity

- For a drug to be useful as either a therapeutic or a scientific tool, it must act selectively on particular cells and tissues. In other words, it must show a high degree of binding-site specificity.

- **Specificity is reciprocal**: individual classes of drug bind only to certain targets, and individual targets recognize only certain classes of drug.

- **No drugs are completely specific in their actions.** In many cases, increasing the dose of a drug will cause it to affect targets other than the principal one, and this can lead to side effects.

Receptor classification

- Based on pharmacological response

- Direct measurement of ligand binding to receptors

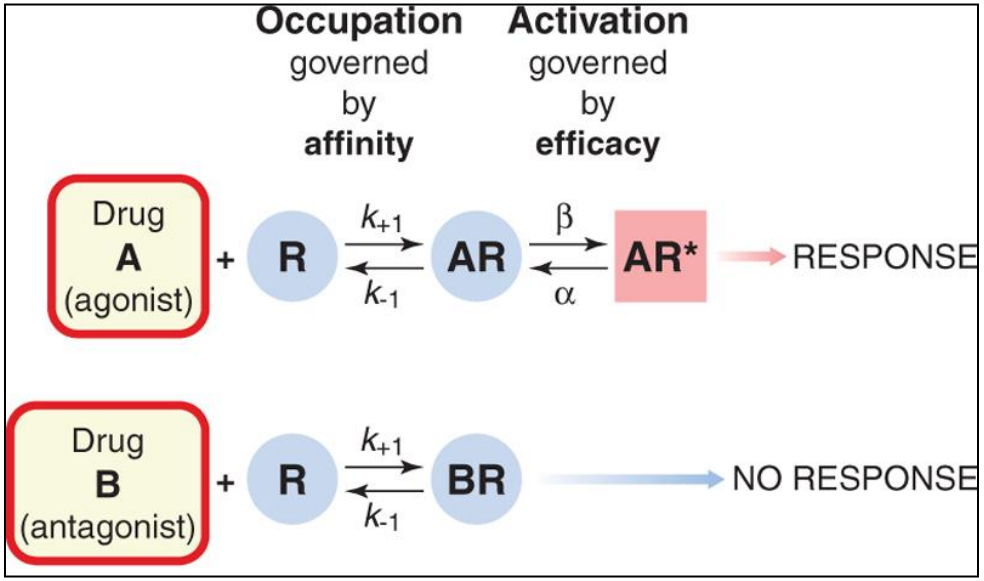
- Molecular cloning

- Analysis of the biochemical pathway that are linked to receptor activation

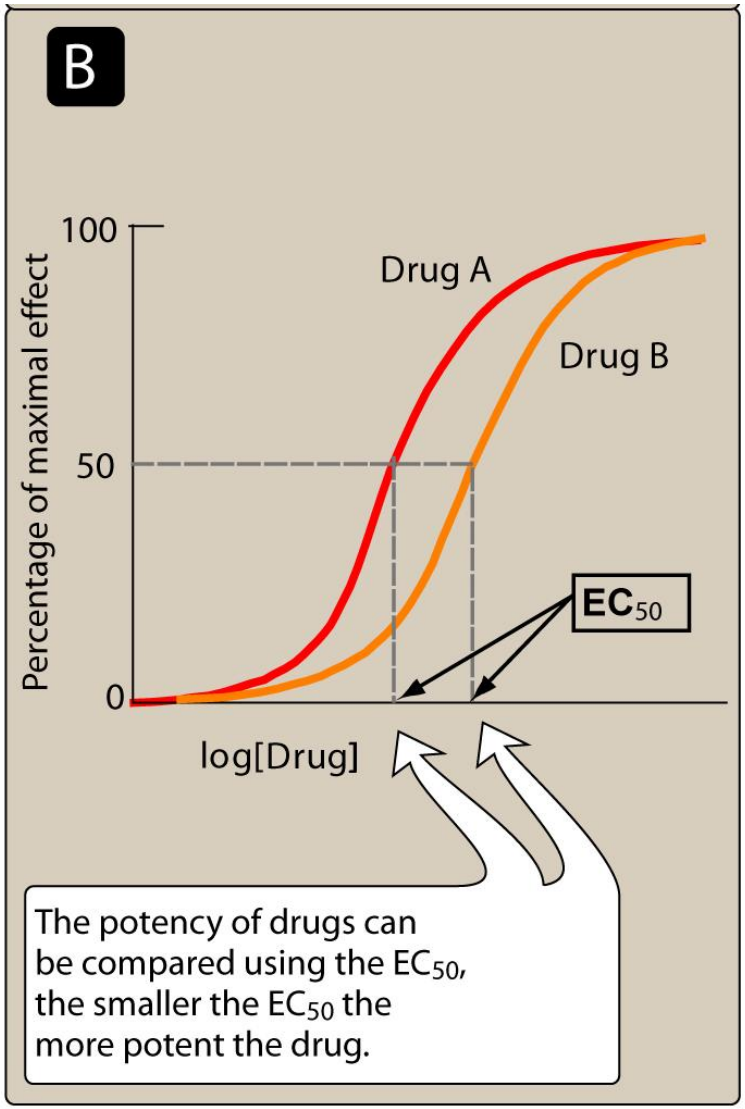
Drug-Receptor Interactions

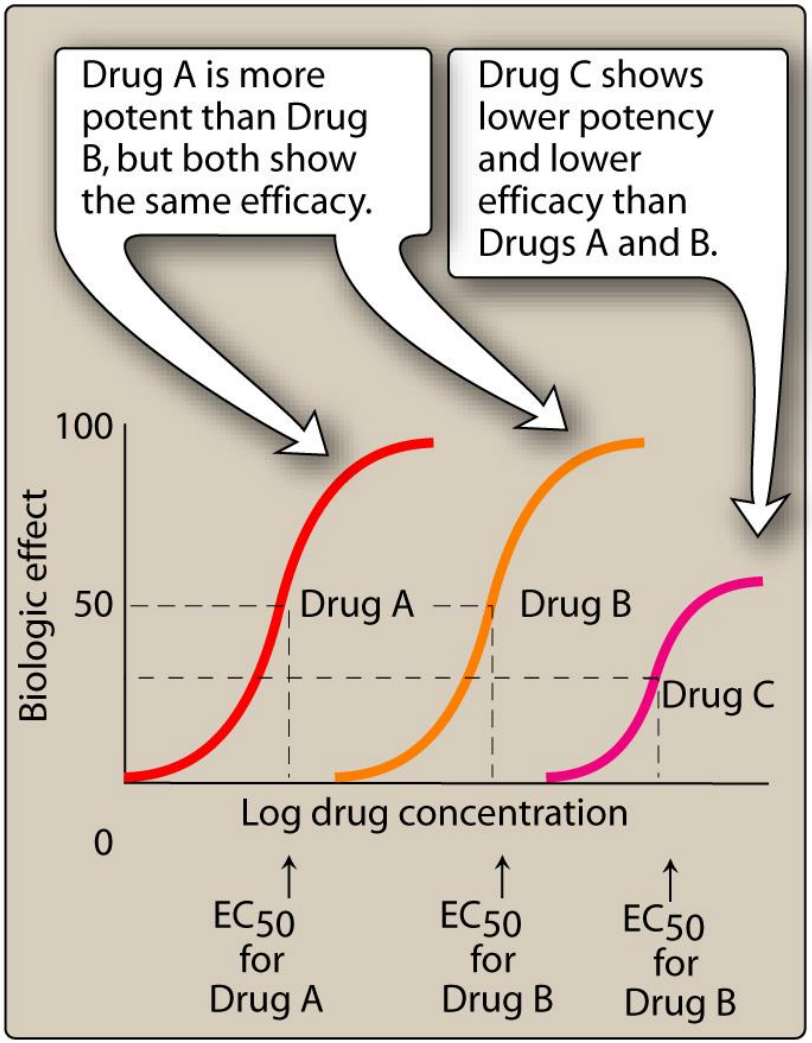
Occupation of a receptor by a drug molecule may or may not result in *activation* of the receptor.

- The tendency of a drug to bind to the receptors is governed by its ***affinity***, whereas the tendency for it, once bound, to activate the receptor is denoted by its ***efficacy***.



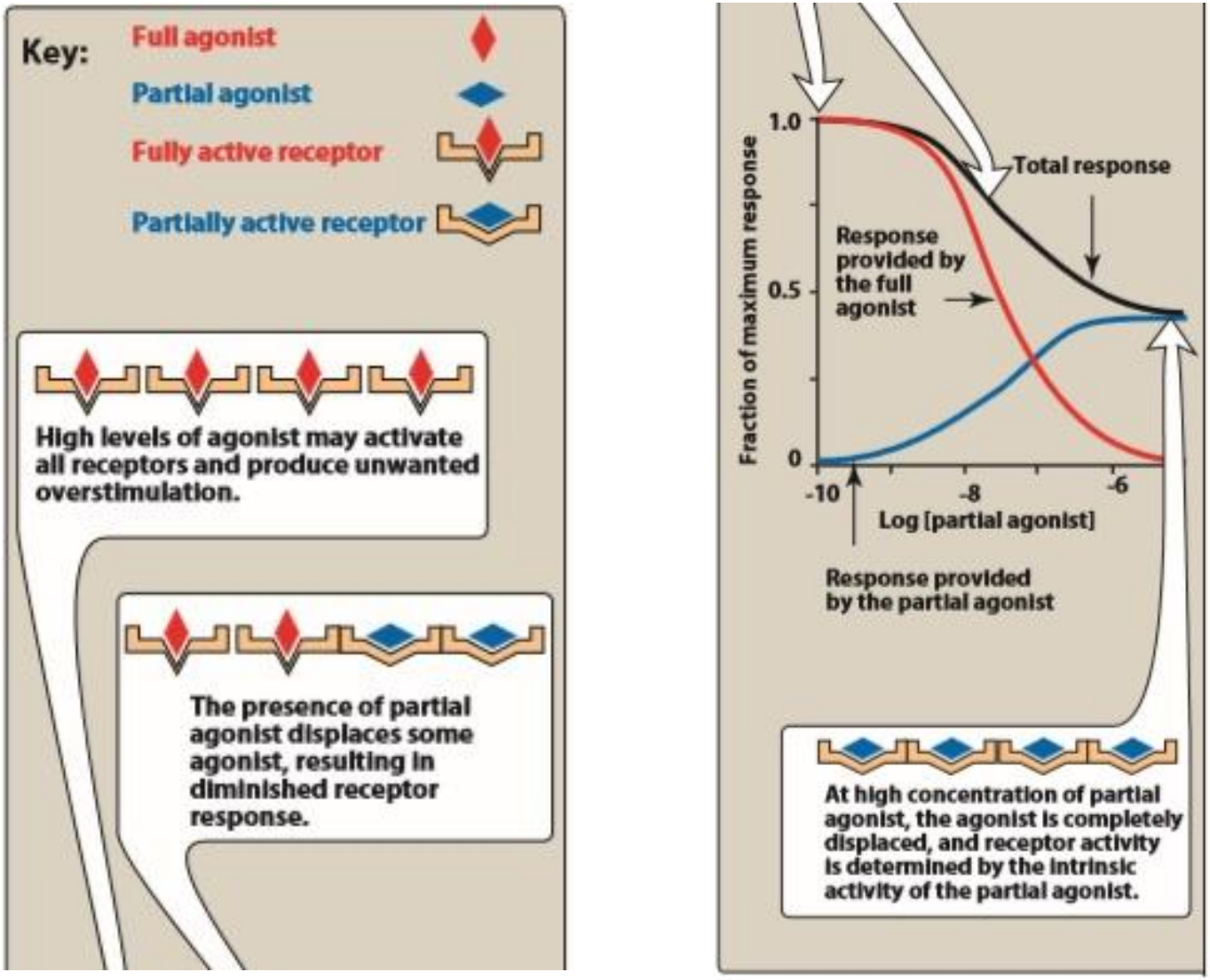
The distinction between drug binding and receptor activation


B
100
50
0
Percentage of maximal effect
Drug A
Drug B
EC50
log[Drug]
The potency of drugs can be compared using the EC50, the smaller the EC50 the more potent the drug.


Drug A is more potent than Drug B, but both show the same efficacy.
Drug C shows lower potency and lower efficacy than Drugs A and B.
100
50
0
Biologic effect
Drug A
Drug B
Drug C
Log drug concentration
EC50 for Drug A
EC50 for Drug B
EC50 for Drug B

Agonists, antagonist and efficacy

- Drugs acting on receptors may be **agonists** or **antagonists.**
- **Agonists** initiate changes in cell function, producing effects of various types; **antagonists** bund to receptors without initiating such changes.
- Agonist potency depends on two parameters: *affinity* (i.e. tendency to bind to receptors) and *efficacy* (i.e. ability, once bound, to initiate changes that lead to effects).
- For antagonists, efficacy is zero.
- **Full agonists** (which can produce maximal effects) have high agonists.
- **Partial agonists** (which can produce only submaximal effects) have intermediate efficacy.



Effects of partial agonists.

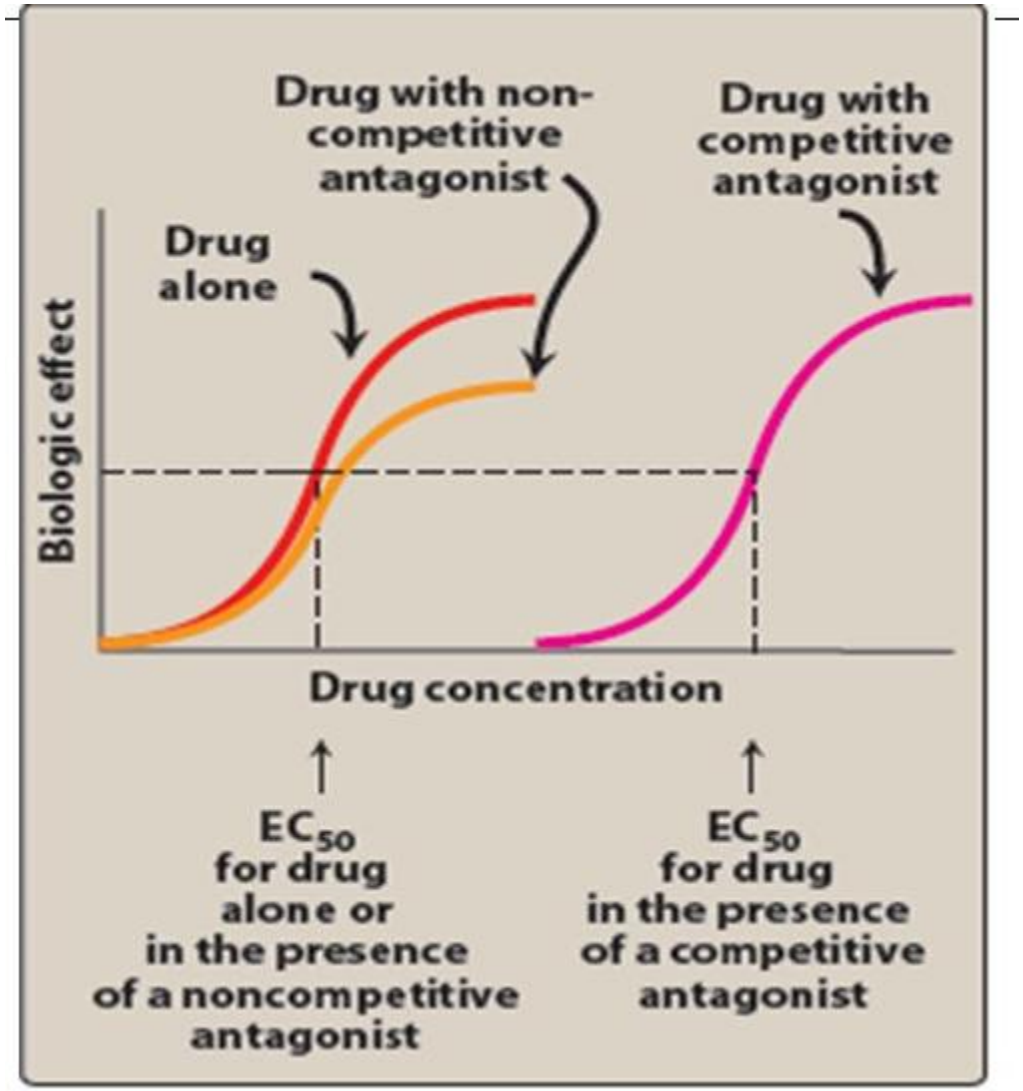
Antagonists

Competitive antagonists:

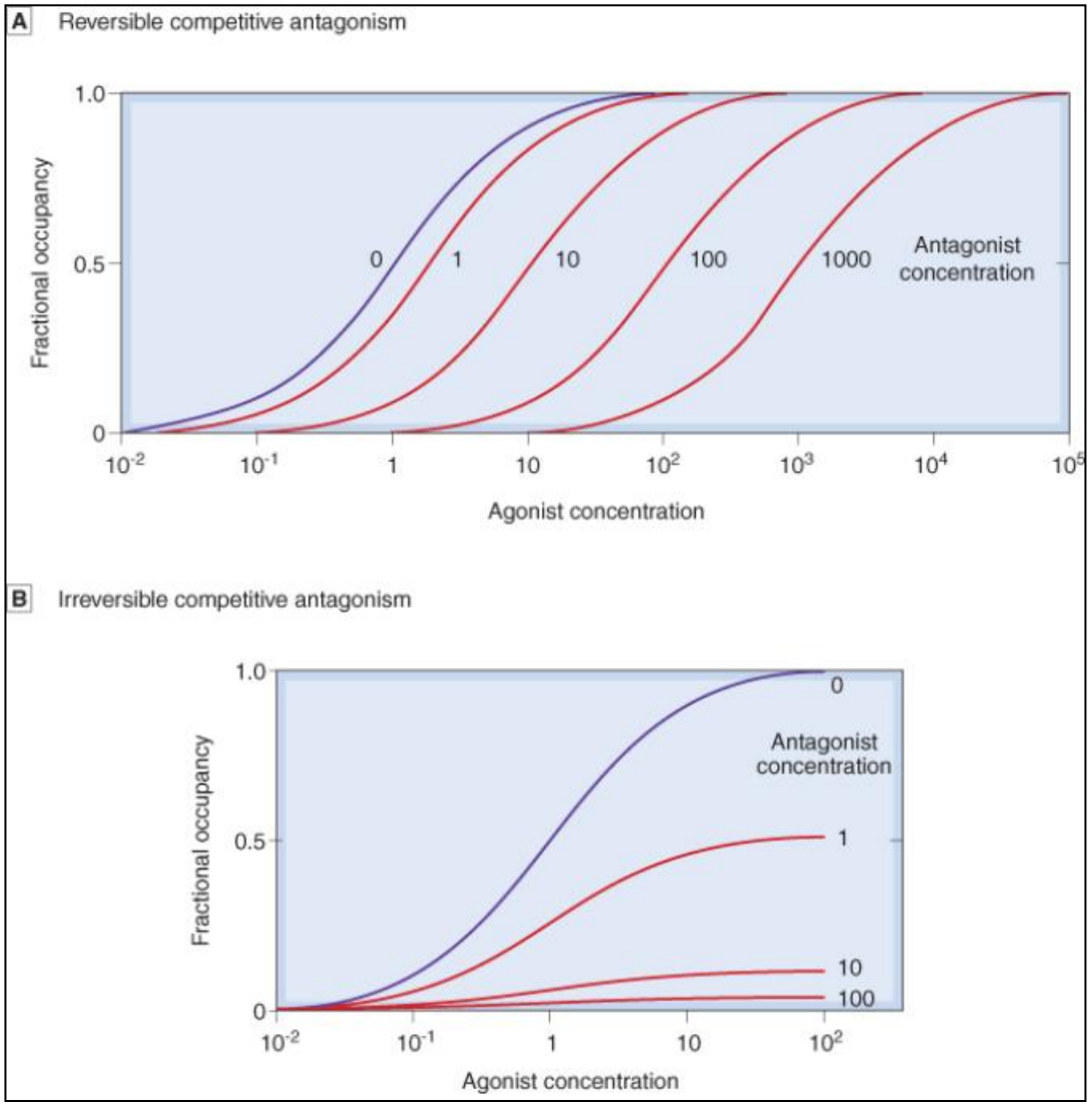
- If the antagonist binds to the same site on the receptor as the agonist in a reversible manner, it is “competitive.”
- In the presence of the antagonist, the agonist log concentration-effect curve is shifted to the right without change in slope or maximum, the extent of the shift being a measure of the dose ratio.

Irreversible antagonists:

- Irreversible antagonists bind covalently to the active site of the receptor, thereby permanently reducing the number of receptors available to the agonist.
- An irreversible antagonist causes a downward shift of the E_{max}, with no shift of EC_{50} values.



Effects of drug antagonists. EC_{50}= drug dose that shows 50% of maximal response.

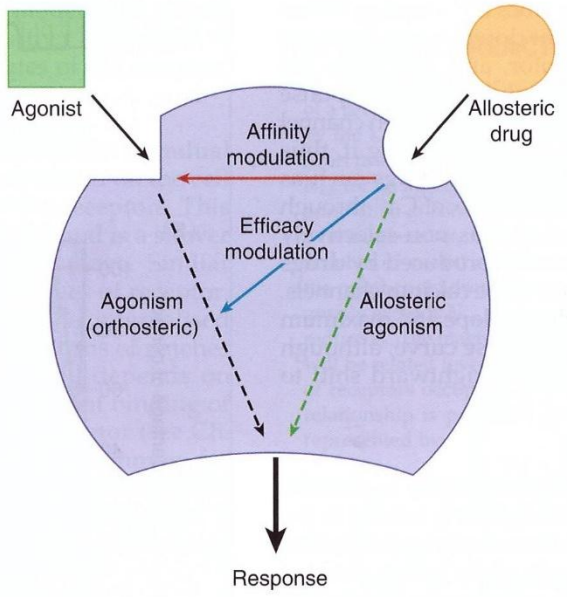


Hypothetical agonist concentration-occupancy curves in the presence of reversible and irreversible competitive antagonists. The concentrations are normalized with respect to the equilibrium constants, K, (i.e. 1.0 corresponds to a concentration equal to K and results in 50% occupancy).

Antagonists

Allosteric antagonists:

- An allosteric antagonist binds to a site (allosteric site) other than the agonist-binding site and prevents receptor activation by the agonists.
- An allosteric antagonist causes a downward shift of the E_{max} of an agonist, with no change in the EC_{50} value.



Types of drug antagonism

- **Chemical antagonism** (interaction in solution)

  : chelating agents (e.g. dimercaprol), neutralising antibody (e.g. infliximab)

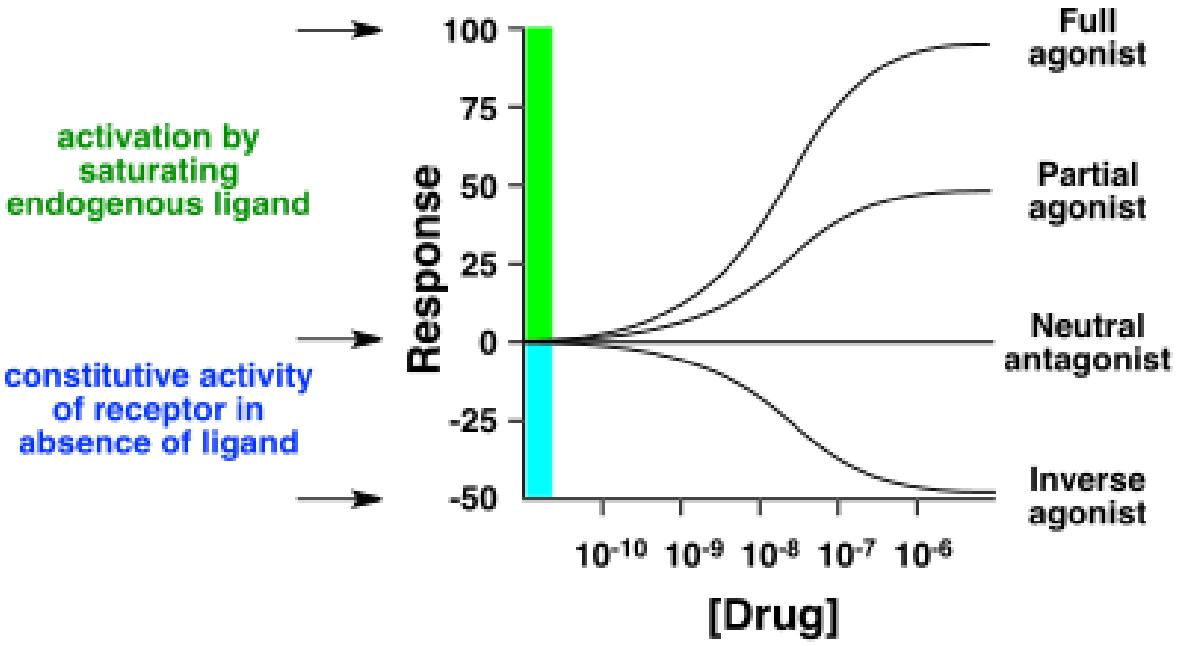
- **Pharmacokinetic antagonism** (one drug affecting the absorption, metabolism or excretion of the other)

  : e.g. cimetidine -- lidocaine

- **Physiological antagonism** (two agents producing opposing physiological effects)

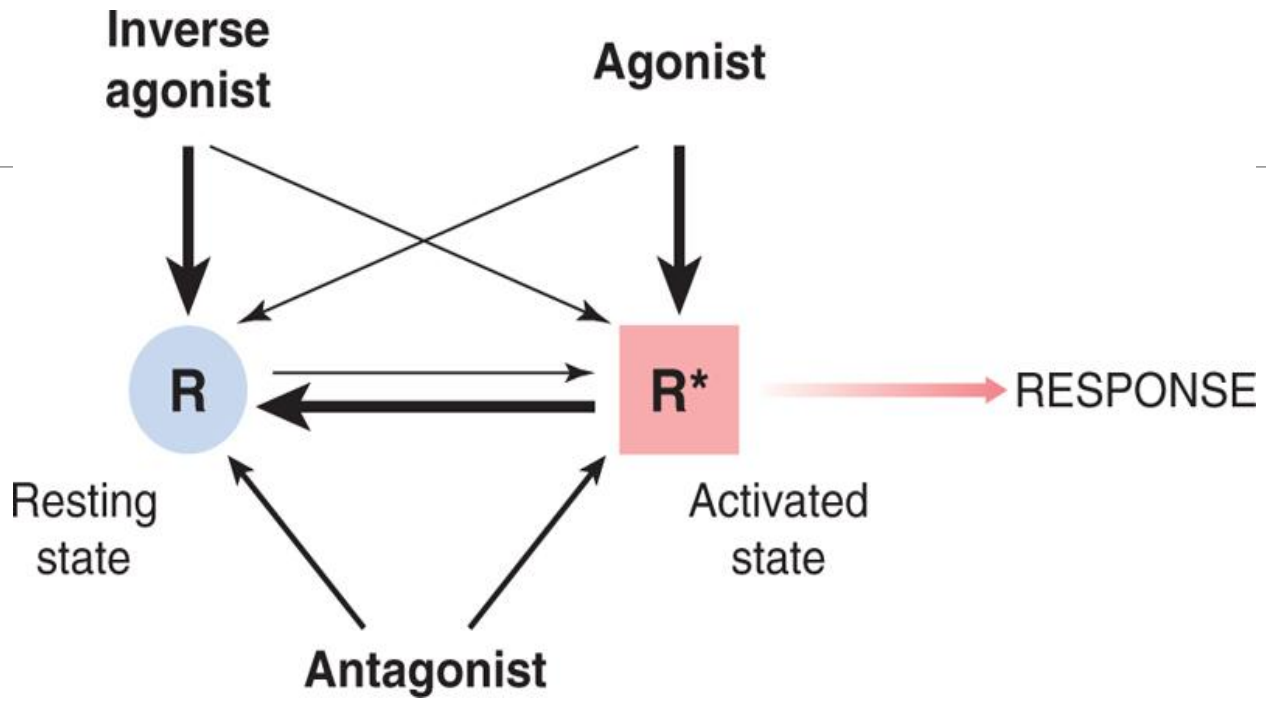
  : e.g. histamine -- epinephrine

Inverse agonist



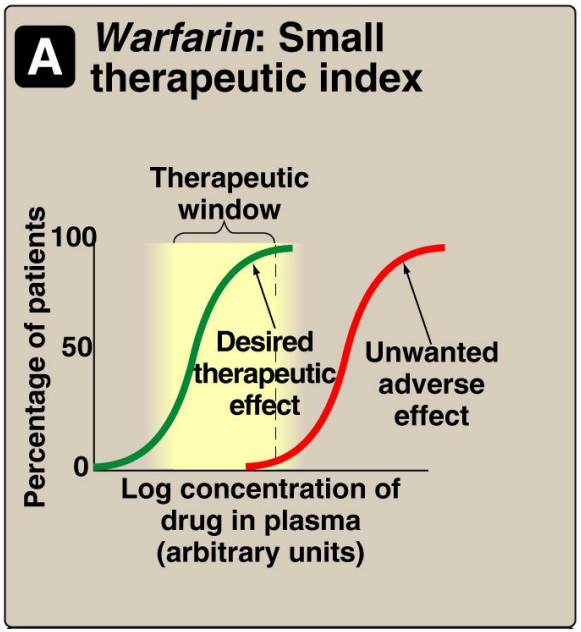
- Inverse agonists show selectivity for the resting state of the receptor, this being of significance only in situations where the receptors show constitutive activity.

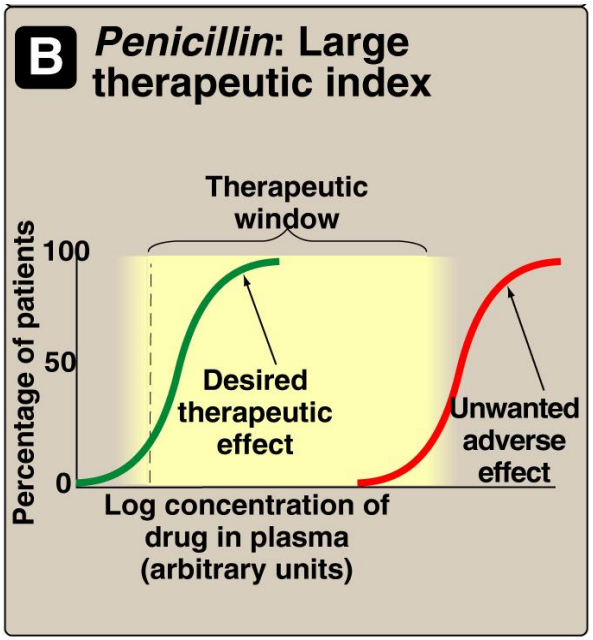
The two-state model



- According to the two-state model, efficacy reflects the relative affinity of the compound for the resting and activated states of the receptor.
- Agonists show selectivity for the activated state; antagonists show no selectivity.

Therapeutic index (TI)





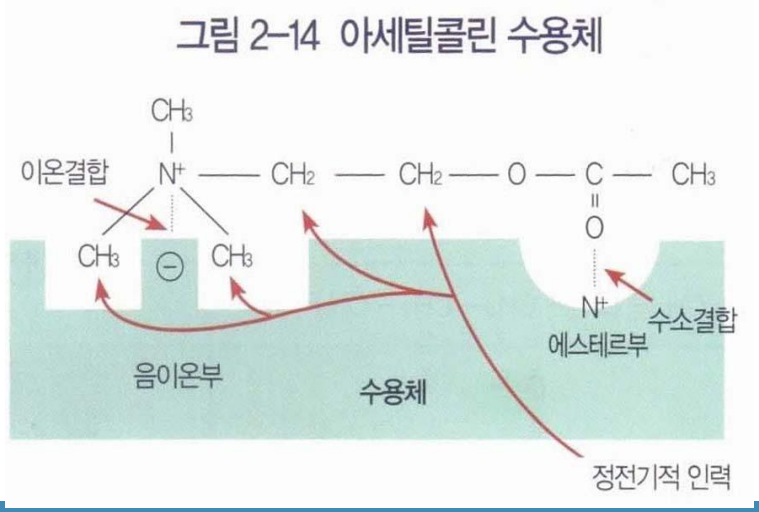
- ED_{50} (median effective dose)
- TD_{50} (median toxic dose)

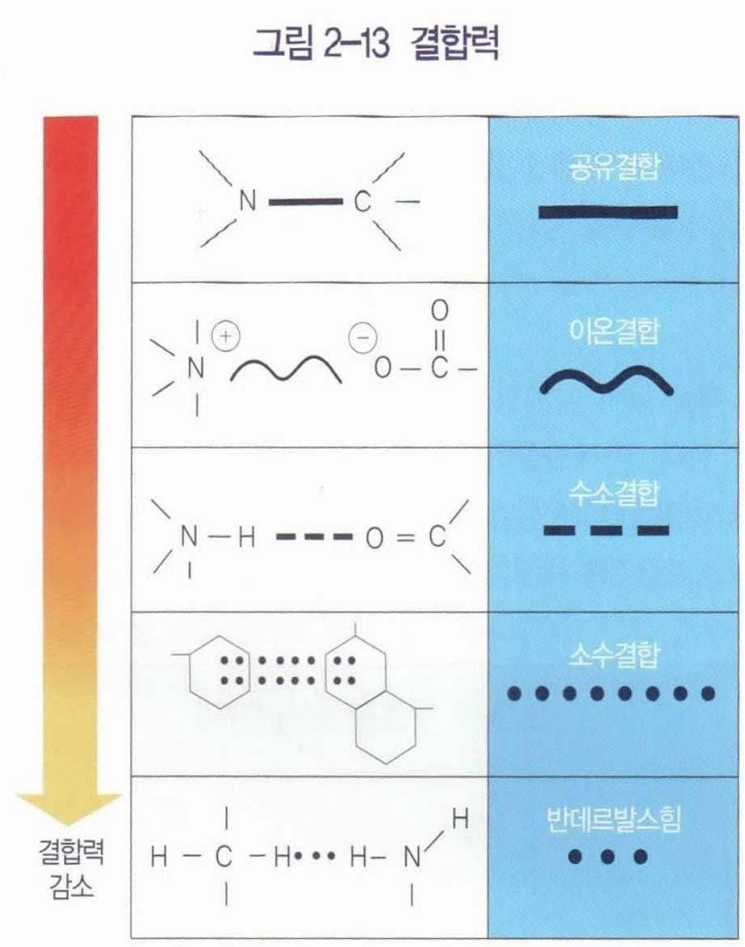
$$TI = TD_{50}/ED_{50}$$

- A large number = a wide margin of safety
- A small number = a small margin of safety
- Safe drug: TI is more than 10 or more

Drug-Binding Forces

- Covalent bonds
- Ionic bonds
- Hydrogen bonds
- Van der Waals forces
- Hydrophobic interactions





Concentration-response relationship

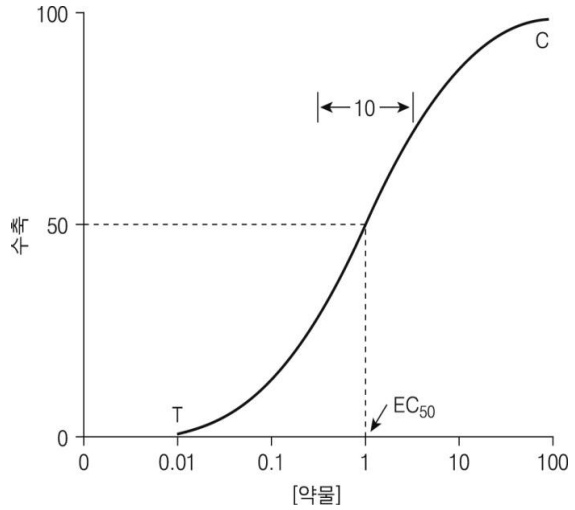
Occupation Concept

- The magnitude of a pharmacologic response elicited by a drug that reversibly combines with its receptor is directly proportional to the number of receptors occupied by the drug.

$$D + R \underset{k_2}{\overset{k_1}{\rightleftharpoons}} DR \longrightarrow \text{Effect}$$

$$\text{Effect} = \frac{E_{max} \times [D]}{K_D + [D]}$$

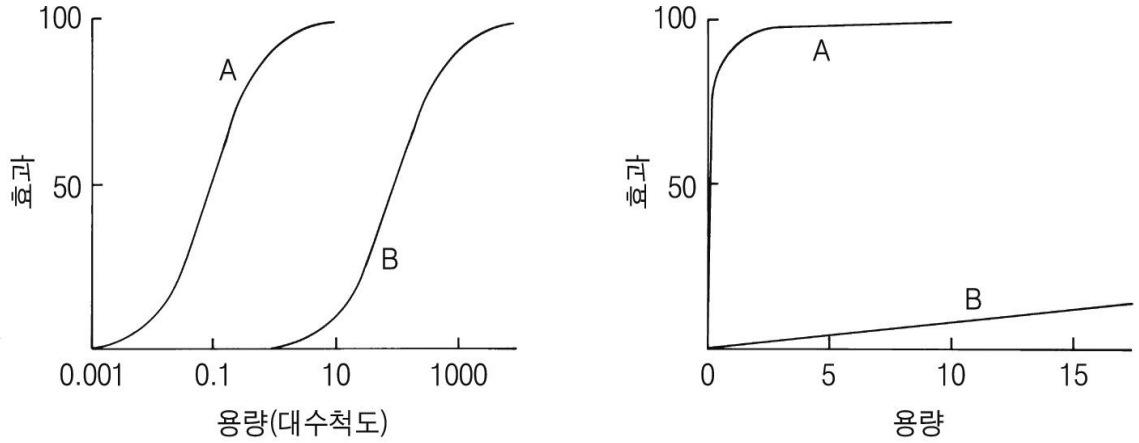
K_D **(the dissociation constant)=k_2/k_1**



Theoretic concentration-response curve (log scale) for a smooth muscle stimulant.

Concentration-response relationship

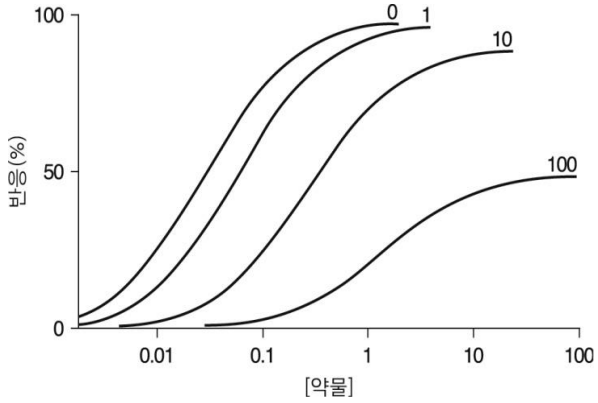
Occupation concept



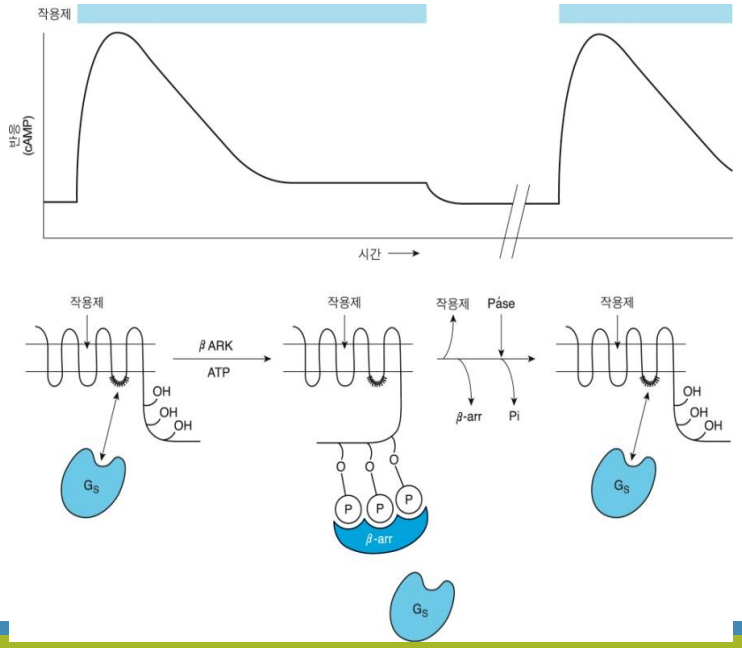
Dose-effect curves for two drugs differing in receptor affinity by a factor of 1000. *Left*, A log scale. *Right*, An arithmetic scale.

- **Limitation of occupation theory**

Spare receptor



- **Regulation of receptor**
  - Desensitization
  - Down-regulation
  - Up-regulation



Drug Effects

- **Drugs act mainly on cellular targets, producing effects at different functional levels (e.g. biochemical, cellular, physiological and structural).**

- **The direct effect of the drug on its target produces acute responses at the biochemical, cellular or physiological levels.**

- **Prolonged receptor activation generally leads to delayed long-term effects, such as desensitization or down-regulation of receptors, hypertrophy, atrophy or remodelling of tissues, tolerance, dependence and addiction.**

- **Long-term delayed responses result from changes in gene expression, although the mechanisms by which the acute effects bring this about are often uncertain.**

- **Therapeutic effects may be based on acute responses (e.g. the use of bronchodilator drugs to treat asthma) or delayed responses (e.g. antidepressants).**

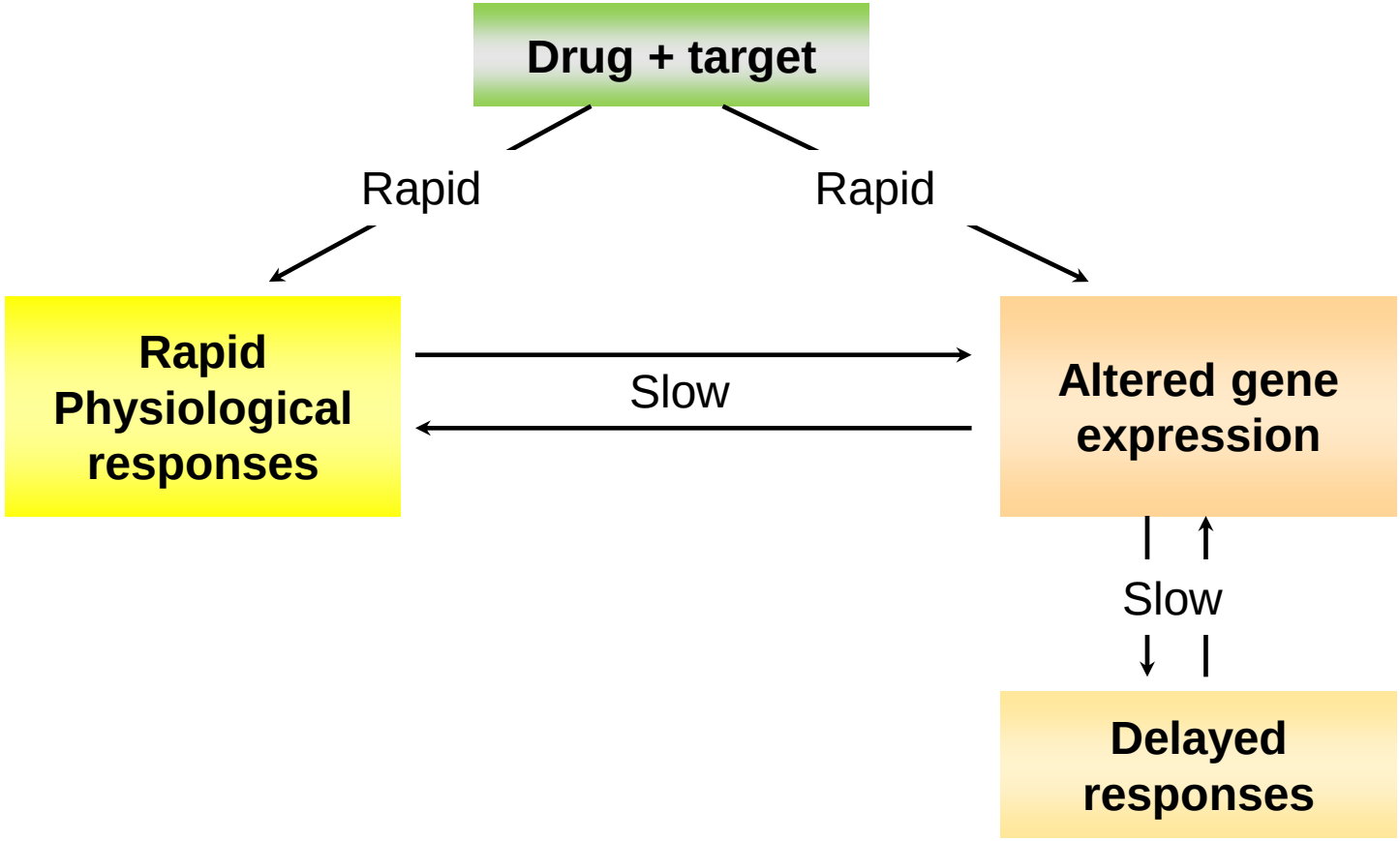


Fig. Early and late responses to drugs.

Many drugs act directly on their targets (left-hand arrow) to produce a rapid physiological response. If this is maintained, it is likely to cause changes in gene expression that give rise to delayed effects. Some drugs (right-hand arrow) have their primary action on gene expression, producing delayed physiological responses. Drugs can also work by both pathways. Note the bidirectional interaction between gene expression and response.